

SOWMYA KANITHI

AI/ML Engineer | Generative AI Engineer | Agentic AI • RAG • AWS Cloud

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SUMMARY

AI/ML Engineer with hands-on experience across Generative AI, Retrieval-Augmented Generation (RAG), and Agentic AI, gained through applied internship work and self-driven project building. Experienced in developing Python-based backend services with FastAPI and building AI agents using AWS Strands Agents SDK, Amazon Bedrock, and Bedrock AgentCore, including deployment and monitoring on AWS. Currently deepening expertise in the Model Context Protocol (MCP) and Azure AI, gained by contributing to HR agent and workflow-automation initiatives. Strong foundation in Machine Learning, Deep Learning, and NLP, with a continuous-learning mindset and a track record of delivering practical, end-to-end AI applications grounded in real-world implementation.

SKILLS

Artificial Intelligence & Generative AI: Machine Learning, Deep Learning, Generative AI, Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), Embeddings, Vector Search, AI Agents, Agentic AI, Agent Orchestration, Tool Calling, Strands Agents SDK, Amazon Bedrock AgentCore, Agent Runtime Architecture, AI Guardrails, AI Safety, Human-in-the-Loop, Multimodal AI, Computer Vision, NLP

Programming Languages & Full Stack Development: Python, JavaScript, SQL, C/C++, HTML5, CSS3, FastAPI, REST APIs, Pydantic, Uvicorn, Mangum, React, Vite, React Router, Tailwind CSS

Cloud Platforms – AWS: Amazon Bedrock, Bedrock Guardrails, Bedrock AgentCore, AWS Lambda, Amazon S3, Amazon DynamoDB, Amazon ECR, AWS CodeBuild, Amazon CloudWatch, AWS IAM, Amazon API Gateway, Amazon OpenSearch, AWS CLI

Currently Learning – MCP & Azure AI: Actively upskilling in the Model Context Protocol (MCP), MCP tool registries, and agent-governance patterns, alongside Azure AI, Azure AI Agents, Azure Portal, Azure CLI, Azure Resource Manager, Resource Groups, Subscriptions & Tenants, and Regions & Availability Zones as part of ongoing enterprise AI workflow exposure

Machine Learning & Deep Learning Frameworks: Scikit-learn, PyTorch, TensorFlow, CNNs, YOLOv8, OpenCV, Hugging Face Transformers, LangChain, FAISS, ChromaDB, NLTK, SpaCy, Pandas, NumPy

Developer Tools & DevOps: Git, GitHub, Docker, Streamlit, Gradio, Hugging Face Spaces, VS Code, Jupyter Notebook, Google Colab

EXPERIENCE

AIML Intern – *Tachyon Technologies – Hyderabad, Telangana (Onsite)* May 2026 – Present

- Contributing to Agentic AI application development in Python — building and integrating AI agents with tool-calling capabilities using AWS Strands Agents SDK and Amazon Bedrock — alongside RAG pipeline components (document ingestion, chunking, embeddings, semantic retrieval) using Bedrock and ChromaDB/OpenSearch.
- Supporting FastAPI backend development (structured logging, health checks, request correlation IDs) and agent deployment to Amazon Bedrock AgentCore Runtime, working with AWS IAM, ECR, CodeBuild, and CloudWatch.
- Working with Amazon Bedrock Guardrails for AI safety and guardrail configuration checks, and contributing to enterprise HR agent workflow analysis (Ask HR, Hiring use cases), including process-flow mapping and Build-vs-Buy evaluation of ticketing integrations (ServiceNow, Freshdesk).
- Building working knowledge of Azure AI and the Model Context Protocol (MCP), including tool registries and agent-governance concepts, as part of ongoing skill development.

Machine Learning Intern – *Talent-Trek e-Learning, Bangalore (Remote)* May 2025 – Aug 2025

- Preprocessed and analyzed datasets using Pandas and NumPy to prepare data for model training.
- Built and evaluated machine learning models achieving 88–95% accuracy.

Machine Learning Intern – *InternshipStudio, India (Remote)* Aug 2024 – Jan 2025

- Performed exploratory data analysis (EDA), data cleaning, and feature selection on structured datasets.
- Built end-to-end ML pipelines, iterating on feature engineering and model selection to improve performance.

PROJECTS

Conference Room Booking Agent

Python, AWS Strands Agents SDK, Amazon Bedrock, Bedrock AgentCore, DynamoDB, IAM, ECR, CodeBuild, CloudWatch

GitHub: github.com/sowmya13531/conference

- Designed and built a multi-agent conference room booking system on the AWS Strands Agents SDK, implementing agent and tool interaction logic in Python, deployed to Amazon Bedrock AgentCore Runtime and backed by Amazon DynamoDB for data persistence.
- Configured AWS IAM, Amazon ECR, and AWS CodeBuild for the deployment pipeline; used Amazon CloudWatch for runtime monitoring and troubleshoot dependency, runtime, and environment issues.

AURA – AI Guardrails

Python, FastAPI, Amazon Bedrock Guardrails, DynamoDB, AWS Lambda, AWS IAM, Amazon CloudWatch

GitHub: *Private repository*

- Worked on AI guardrail functionality, integrating FastAPI backend APIs with Amazon Bedrock Guardrails for content and safety checks, with API request validation and health-check endpoints.
- Implemented structured logging and request correlation IDs for observability; managed configuration across AWS services (Lambda, DynamoDB, IAM, CloudWatch) and troubleshoot AWS, API, and runtime issues.

TRUTHX – Agentic Multimodal Digital-Media Forensics Console

Multimodal AI, Computer Vision, Audio Analysis, Agentic AI

GitHub: github.com/sowmya13531/TRUTHX

- Explored an agentic, multimodal architecture for digital-media forensics, fusing vision, audio, lip-sync, metadata, and provenance signals into a combined risk assessment.
- Worked on components spanning computer vision and audio analysis for media authenticity checks, and metadata/provenance review to support forgery-risk evaluation.

Mini-OpenWorker – Agentic AI Runtime & Tool Orchestration

Python, Agentic AI, LLMs, Tool Orchestration, Agent Governance

GitHub: github.com/sowmya13531/Openworker-mini

- Built and explored a lightweight Mini-OpenWorker implementation in Python to understand core agentic AI architecture, including the agent execution loop, LLM-driven tool calling, and a modular tool registry for orchestrating multi-step agent tasks.
- Implemented agent-governance patterns — including risk controls, security checks, and human-in-the-loop approval workflows — to reinforce safe, controllable agentic AI design; MCP integration planned as a next step.

EDUCATION

TiHAN, IIT Hyderabad x Masai – Hyderabad, Telangana

AI & ML Course with Drone Technologies

Jul 2025 – Feb 2026

Rajiv Gandhi University of Knowledge Technologies, IIIT Ongole – Prakasam, Andhra Pradesh

B.Tech in Electronics and Communication Engineering, CGPA: 8.6

Jan 2022 – May 2026

Rajiv Gandhi University of Knowledge Technologies, IIIT Ongole – Prakasam, Andhra Pradesh

Pre-University Course (PUC – MPC), CGPA: 9.0

Jan 2020 – May 2022

CERTIFICATIONS

- AWS Technical Essentials – Amazon Web Services ([View Credential](#))
- Specialized Program in AI & ML – IIT Hyderabad ([View Credential](#))